

Tips and hints

Throughout the book you'll find coloured boxes offering extra information and useful, practical advice and tips.

Purple **IN DEPTH** boxes go further into a topic, giving tips or noteworthy information on it.

IN DEPTH

Steganography

Steganography involves sending secret messages by hiding the fact that there is a message. This is like messages written in invisible ink. Information can be hidden in a digital image by using a program that changes only one bit of each binary number that represents the colours of the pixels. The same program can also extract a hidden message.

TOP TECH

Trending tools

More than 600 million tweets are posted every day. By analysing them as they happen, Twitter is able to identify and highlight trends in what is being discussed. People are often alerted to an event when it first starts trending on Twitter. Similarly, the Google Trends tool gives users access to data on Google

searches. It's possible to see graphs of how often people across the world searched for a particular topic, or top 10 lists of popular searches in different categories. Both tools give users a picture of how people in their own vicinity and other countries are reacting to events.

Orange boxes are **TOP TECH**. They explain some of the latest developments within computing.

LINGO

Communication protocols

When two devices communicate, a protocol dictates whose turn it is to send data, what kind of data is being sent, and how this data is formatted.

Protocol: A set of rules that governs the transmission of data between devices.

HTTP (Hypertext Transfer Protocol): Used for visiting webpages.

HTTPS (Hypertext Transfer Protocol Secure): A secure HTTP.

DHCP (Dynamic Host Configuration Protocol): All computers use this to obtain their IP address from a router.

Yellow colour boxes explain **LINGO**: terms that might not be clear to most readers.

BIOGRAPHY boxes are green, and give background on the lives and ideas of important people in computer science.

BIOGRAPHY

Ada Lovelace

English mathematician Ada Lovelace (1815–1852) created the world's first computer program when she wrote the earliest algorithm to be processed on English scientist Charles Babbage's (1791–1871) Analytical Engine machine in 1844. She was the first to see that computers could do so much more than basic number-crunching and calculations.

REAL WORLD

First integrated circuit

The first integrated circuit was created in 1958 by American electrical engineer Jack Kilby (1923–2005). Before Kilby's invention, machines used vacuum tubes, which were bulky and unreliable. Kilby's IC was based on tiny transistors, and all the parts were made on the one piece of material: the integrated chip was born.

BLUE boxes explain how an idea in the computer science world worked, or works, in the **REAL WORLD**.

Code boxes

Though there are no exercises in this book, some pages feature snippets of computer code. These are clearly marked out in grey and blue boxes. These feature on every entry in the "Programming languages" chapter, through a simple "Hello, World!" program. A "Hello, World!" program is the simplest way to demonstrate the syntax of a programming language, and is usually the first working program that someone unfamiliar with a new programming language attempts.

```
#include <stdio.h>

int main()
{
    int i;
    for (i = 0; i < 5; i++){
        printf("Hello, World!");
    }
    return 0;
}
```